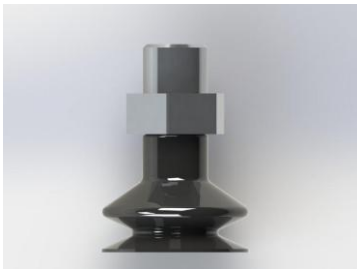
# Suction Cup Actuator Sizing

The suction cup shown in the figure is used as an end effector for the cartesian robot. It is compatible with a Φ6 mm pneumatic tube and is utilized to lift the product's parts, which are made from Artelone and weigh approximately 32.52 grams (0.03252 kg).



## Suction Force Calculations ($F_{\text{suction}}$)

To calculate the required linear acceleration ($a$), we apply the motion relationship:

$$a = \frac{L \cdot \ddot{\theta}}{2\pi} \text{ and } a = \ddot{\theta}\, r$$

Given Parameters:

- Mass ($m$) = 32.52 g $= 0.03252$ kg
- Gravitational Acceleration ($g$) = 9.81 m/s$^2$
- Weight ($W = m \cdot g$) = 0.03252 kg × 9.81 m/s$^2$ $= 0.31902$ N
- Z-Axis Power Screw Lead ($L$) = 8 mm $= 0.008$ m
- Z-Axis Angular Acceleration ($\ddot{\theta}_z$) = 81.50 rad/s$^2$
- Y-Axis Pulley's Pitch Radius: $r = 9.55\ mm$
- Y-Axis Angular Acceleration ($\ddot{\theta}_y$) = 11.35 rad/s$^2$

Linear Vertical Acceleration ($a_z$):

$$a_z = \frac{0.008 \text{ m} \times 81.50 \text{ rad/s}^2}{2\pi} = \frac{0.652}{2\pi} \approx 0.10377 \text{ m/s}^2$$

Vertical Acceleration Force ($F_{a_z}$):

$$F_{a_z} = m \cdot a_z = 0.03252 \text{ kg} \times 0.10377 \text{ m/s}^2 \approx 0.00337 \text{ N}$$

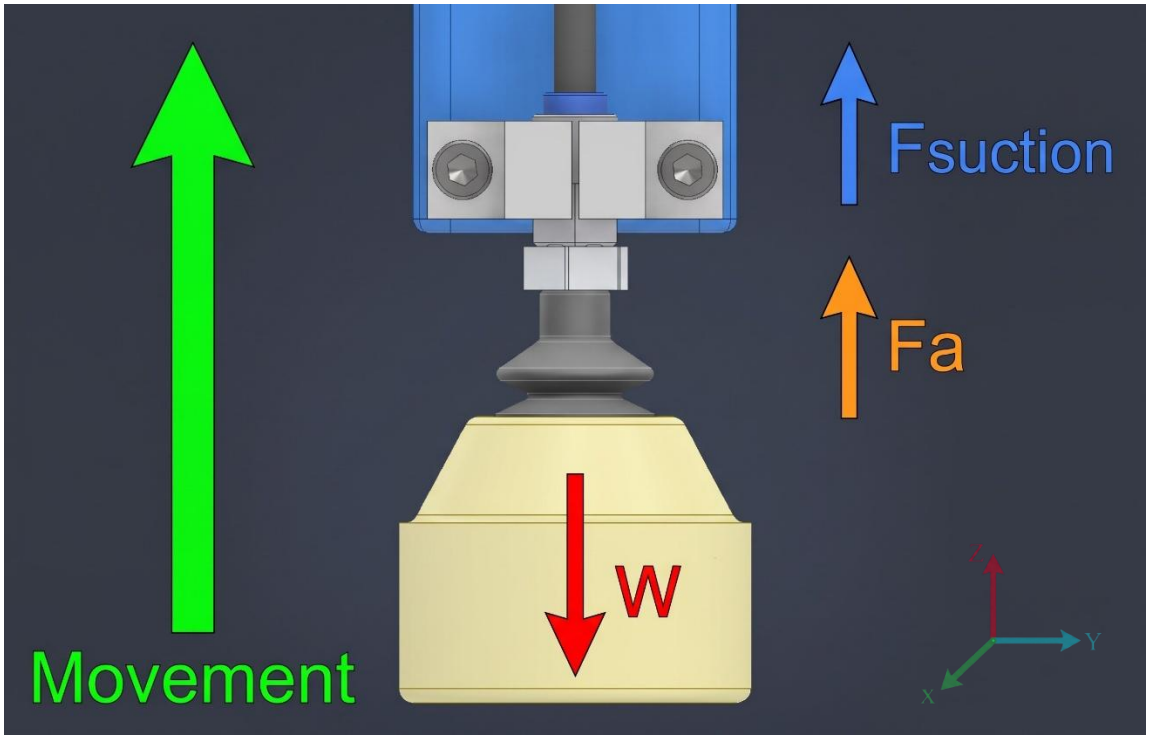
Linear Horizontal Acceleration ($a_y$):

$$a_y = 11.35 \text{ rad/s}^2 \times 0.00955 \text{ m} = 0.1084 \text{ m/s}^2$$

Horizontal Acceleration Force ($F_{a_y}$):

$$F_{a_y} = m \cdot a_y = 0.03252 \text{ kg} \times 0.1084 \text{ m/s}^2 \approx 0.00353 \text{ N}$$

**Case 1:** Upward Movement




Equilibrium of Forces ($\Sigma F_y = 0$):

$$F_{\text{suction}} + F_{a_z} - W = 0$$

Moving terms to one side to establish the suction force requirement:

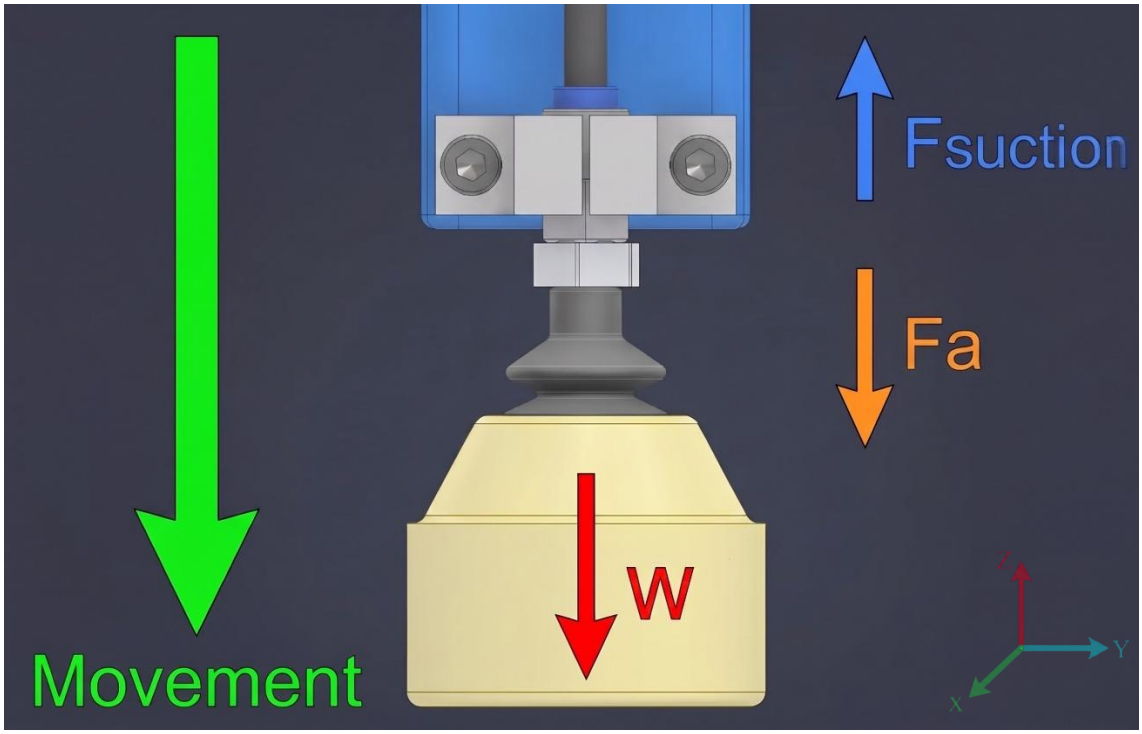
$$F_{\text{suction}} > W - F_{a_z}$$

Calculation**:**

$$F_{\text{suction}} > 0.31902\ \text{N} + 0.00337\ \text{N}$$

$$F_{\text{suction}} > 0.31565\ \text{N}$$

**Case 2:** Downward Movement




Equilibrium of Forces ($\Sigma F_z = 0$):

$$F_{\text{suction}} - F_{a_z} - W = 0$$

Moving terms to one side to establish the suction force requirement:

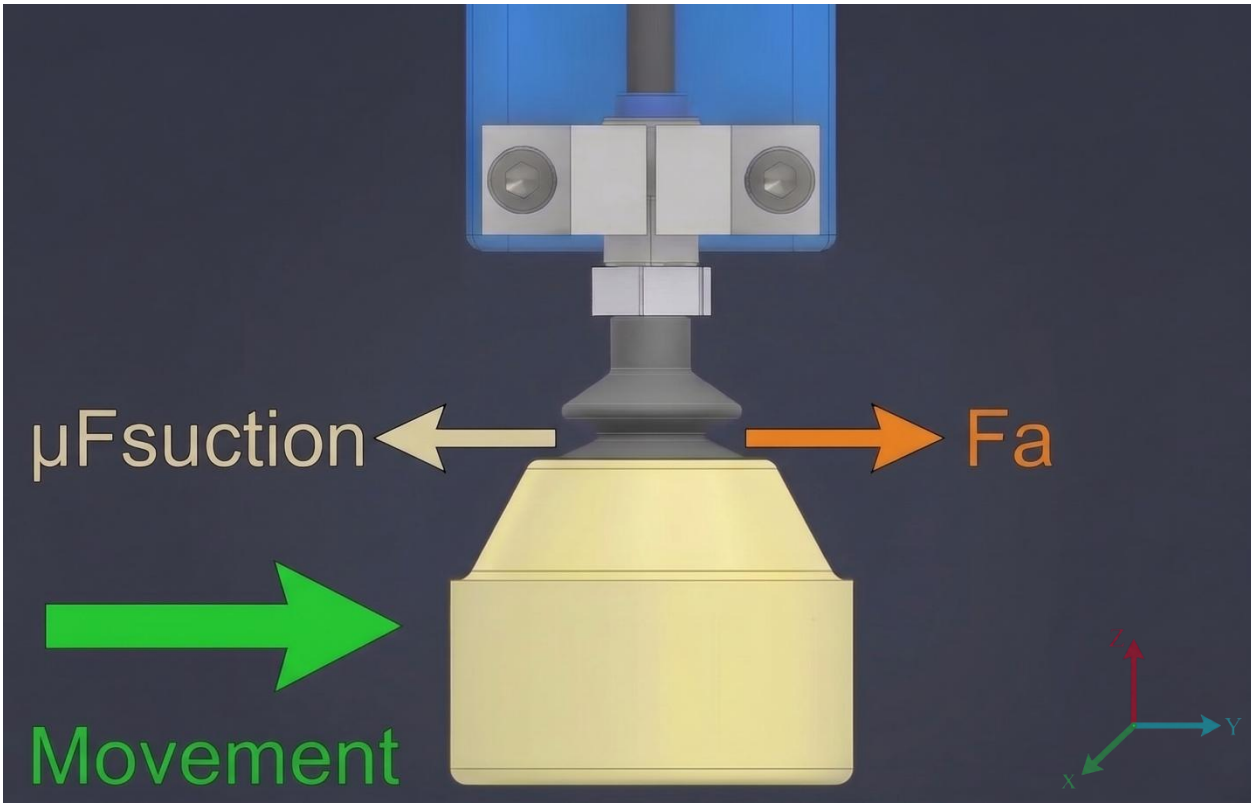
$$F_{\text{suction}} > W + F_{a_z}$$

Calculation:

$$F_{\text{suction}} > 0.31902\ \text{N} + 0.00337\ \text{N}$$

$$F_{\text{suction}} > 0.32239\ \text{N}$$

**Case 3:** Sideway Movement




Equilibrium of Forces ($\Sigma F_y = 0$):

$$F_{a_y} - \mu F_{\text{suction}} = 0$$

Moving terms to one side to establish the suction force requirement:

$$F_{\text{suction}} > \frac{F_{a_y}}{\mu}$$

Calculation:

$$F_{\text{suction}} > \frac{0.00353}{0.5} N$$

$$F_{\text{suction}} > 0.00706 \text{ N}$$

## Suction Force Sizing Criteria

To ensure secure part retention across all operational phases, the suction cup is sized based on the maximum calculated requirement, which occurs during downward movement (**Case 2:** $F_{\text{suction, min}} = 0.32239$ N).

Applying a design safety factor of $n = 2$ to account for surface roughness, dynamic acceleration spikes and vacuum pressure fluctuations, the minimum required suction force is:

$$F_{\text{suction, required}} = F_{\text{suction, min}} \times n = 0.32239 \text{ N} \times 2 = \mathbf{0.64478} \text{ N}$$

## Pneumatic Vacuum System Sizing & Selection

1. Required Vacuum Pressure Calculation:
To achieve the required holding force ($F_{\text{suction}} = 0.645$ N) using the suction tubing ($Inner\ Diameter = 5$ mm, cross-sectional area $A = 19.64$ mm$^2$), the required operating vacuum pressure is calculated as:

$$P_{\text{vacuum}} = \frac{F_{\text{suction}}}{A} = \frac{0.64478 \text{ N}}{1.9635 \times 10^{-5} \text{ m}^2} = 32{,}838 \text{ Pa} \approx -0.328 \text{ bar} \quad (-32.84 \text{ kPa})$$

2. Vacuum Generator Selection & Verification:
To generate this vacuum pressure, a Vacuum Generator "VEV-06HS Venturi Vacuum Generator ($1/8''$ Port)" is selected:



- Supply Pressure Range: $0.1 - 1.6$ MPa ($1.0 - 16.0$ bar)
- Operating Supply Pressure: $5.0 - 8.0$ bar ($0.5 - 0.8$ MPa)
- Max Vacuum Generated: $-0.085$ MPa ($-0.85$ bar)

3. Verification & Safety Margin: Operating the compressed air line at 8.0 bar, the VEV-06HS generator easily delivers up to $-0.85$ bar of vacuum depth. Since the system only requires $-0.328$ bar, the generator operates with a $2.6\times$ pressure safety margin, guaranteeing rapid suction under different conditions.

**Coefficient of Friction value reference:** https://www.schmalz.com/en/support/know-how/vacuum-knowledge/the-vacuum-system-and-its-components/system-design-calculation-example/theoretical-holding-force-of-a-suction-cup